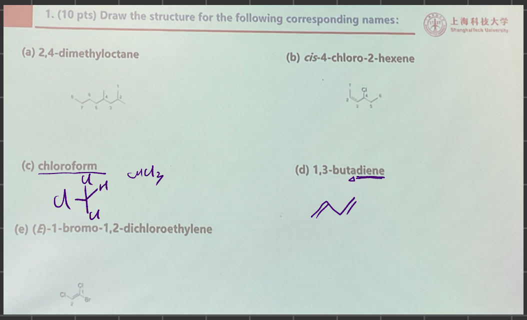


1. (10 pts) Draw the structure for the following corresponding names:

(a) 2,4-dimethyloctane

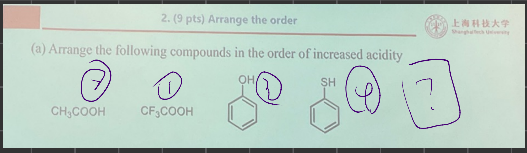
(b) *cis*-4-chloro-2-hexene

(c) chloroform

$CHCl_3$

(d) 1,3-butadiene

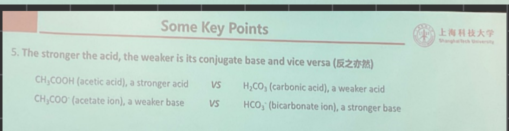
(e) (*E*)-1-bromo-1,2-dichloroethylene



2. (9 pts) Arrange the order

(a) Arrange the following compounds in the order of increased acidity

CH_3COOH ③ CF_3COOH ① OH ② SH ④

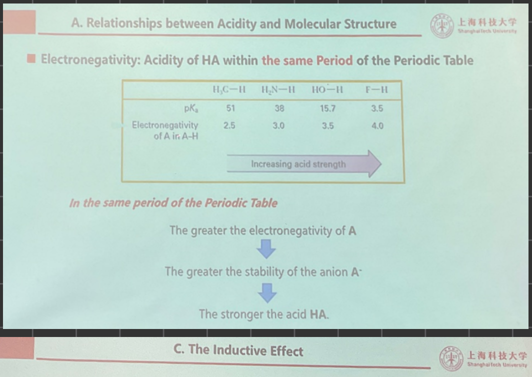


Some Key Points

5. The stronger the acid, the weaker is its conjugate base and vice versa (反之亦然)

CH_3COOH (acetic acid), a stronger acid *VS* H_2CO_3 (carbonic acid), a weaker acid

CH_3COO^- (acetate ion), a weaker base *VS* HCO_3^- (bicarbonate ion), a stronger base



A. Relationships between Acidity and Molecular Structure

- **Electronegativity: Acidity of HA within the same Period of the Periodic Table**

	$H_3C—H$	$H_2N—H$	$HO—H$	$F—H$
pK_a	51	38	15.7	3.5
Electronegativity of A in A–H	2.5	3.0	3.5	4.0

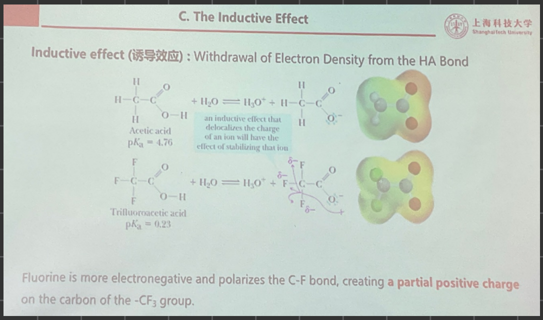
Increasing acid strength

In the same period of the Periodic Table

The greater the electronegativity of A

The greater the stability of the anion A^-

The stronger the acid HA.

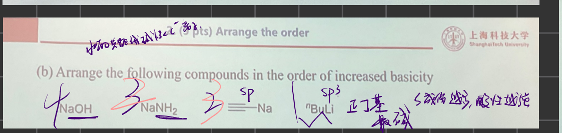


C. The Inductive Effect

Inductive effect (诱导效应): Withdrawal of Electron Density from the HA Bond

+ H2O ⇌ H3O+ +
Acetic acid
pKa = 4.76
an inductive effect that delocalizes the charge of an ion will have the effect of stabilizing that ion
Trifluoroacetic acid
pKa = 0.23
δ−
δ+

Fluorine is more electronegative and polarizes the C-F bond, creating **a partial positive charge** on the carbon of the $-CF_3$ group.

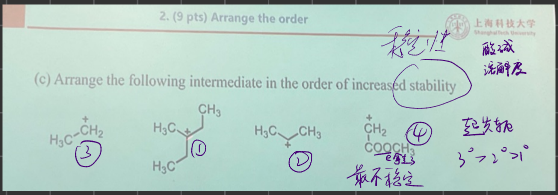


2. (9 pts) Arrange the order

(b) Arrange the following compounds in the order of increased basicity

4 NaOH 3 $NaNH_2$ 2 sp ≡—Na 1 sp³ ^nBuLi 正丁基 锂 碱

S成分越多，酸性越强

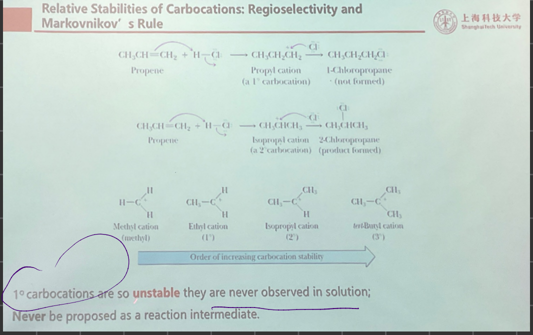


2. (9 pts) Arrange the order

(c) Arrange the following intermediate in the order of increased stability

稳定性 酸碱 溶解度

H3C–CH2+ ③
H3C, CH3, H3C ①
H3C CH3 ②
CH2+ COOCH3 ④
最不稳定
3° > 2° > 1°

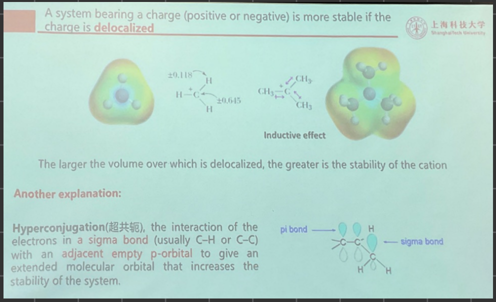


Relative Stabilities of Carbocations: Regioselectivity and Markovnikov's Rule

Propene
Propyl cation (a 1° carbocation)
1-Chloropropane (not formed)
Propene
Isopropyl cation (a 2° carbocation)
2-Chloropropane (product formed)
Methyl cation (methyl)
Ethyl cation (1°)
Isopropyl cation (2°)
tert-Butyl cation (3°)
Order of increasing carbocation stability

1° carbocations are so **unstable** they are never observed in solution;

Never be proposed as a reaction intermediate.



A system bearing a charge (positive or negative) is more stable if the charge is delocalized

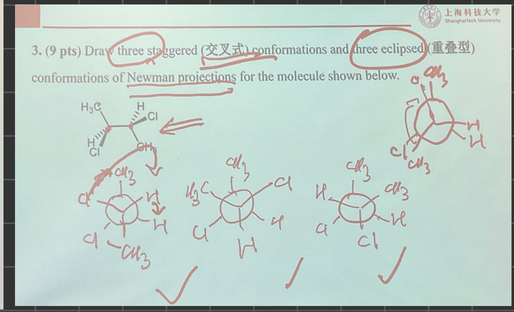
±0.118
±0.645
Inductive effect

The larger the volume over which is delocalized, the greater is the stability of the cation

Another explanation:

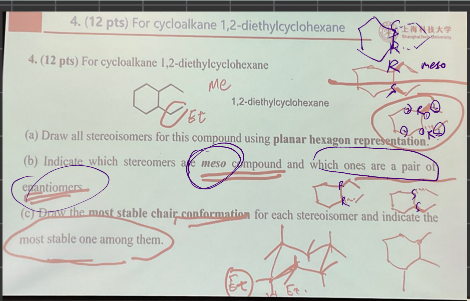
Hyperconjugation(超共轭), the interaction of the electrons in a sigma bond (usually C–H or C–C) with an adjacent empty p-orbital to give an extended molecular orbital that increases the stability of the system.

pi bond
sigma bond



3. (9 pts) Draw three staggered (交叉式) conformations and three eclipsed (重叠型) conformations of Newman projections for the molecule shown below.

H3C
H
Cl
H
Cl
CH3



4. (12 pts) For cycloalkane 1,2-diethylcyclohexane

4. (12 pts) For cycloalkane 1,2-diethylcyclohexane

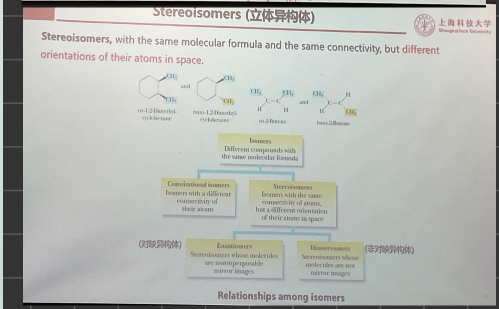
1,2-diethylcyclohexane

Me Et meso

(a) Draw all stereoisomers for this compound using **planar hexagon representation**.

(b) Indicate which stereomers are ***meso*** compound and which ones are a pair of enantiomers

(c) Draw the **most stable chair conformation** for each stereoisomer and indicate the most stable one among them.

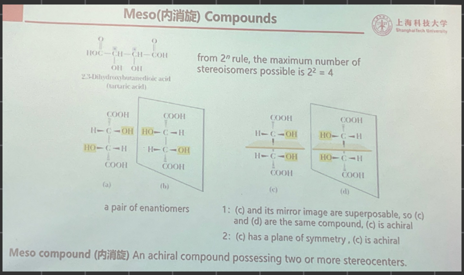


Stereoisomers (立体异构体)

Stereoisomers, with the same molecular formula and the same connectivity, but different orientations of their atoms in space.

cis-1,2-Dimethylcyclohexane
and
trans-1,2-Dimethylcyclohexane
cis-2-Butene
and
trans-2-Butene
Isomers
Different compounds with the same molecular formula
Constitutional isomers
Isomers with a different connectivity of their atoms
Stereoisomers
Isomers with the same connectivity of atoms, but a different orientation of their atoms in space
(对映异构体)
Enantiomers
Stereoisomers whose molecules are nonsuperposable mirror images
Diastereomers
Stereoisomers whose molecules are not mirror images
(非对映异构体)

Relationships among isomers



Meso(内消旋) Compounds

2,3-Dihydroxybutanedioic acid (tartaric acid)
(a)
(b)
(c)
(d)

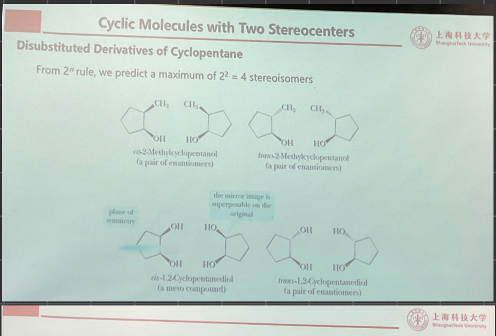
from 2^n rule, the maximum number of stereoisomers possible is $2^2 = 4$

a pair of enantiomers

1: (c) and its mirror image are superposable, so (c) and (d) are the same compound, (c) is achiral

2: (c) has a plane of symmetry , (c) is achiral

Meso compound (内消旋) An achiral compound possessing two or more stereocenters.

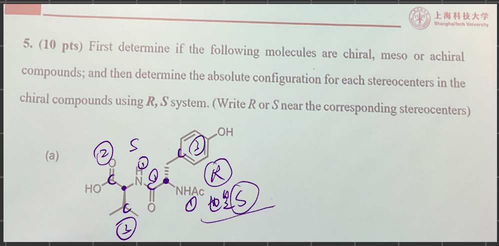


Cyclic Molecules with Two Stereocenters

Disubstituted Derivatives of Cyclopentane

From 2^n rule, we predict a maximum of $2^2 = 4$ stereoisomers

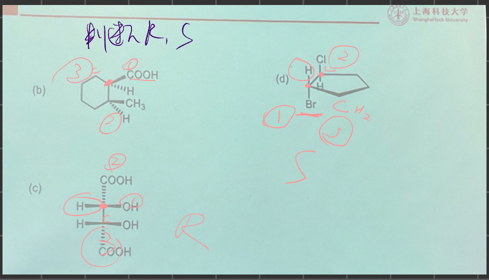
cis-2-Methylcyclopentanol (a pair of enantiomers)
trans-2-Methylcyclopentanol (a pair of enantiomers)
plane of symmetry
the mirror image is superposable on the original
cis-1,2-Cyclopentanediol (a meso compound)
trans-1,2-Cyclopentanediol (a pair of enantiomers)



5. (10 pts) First determine if the following molecules are chiral, meso or achiral compounds; and then determine the absolute configuration for each stereocenters in the chiral compounds using ***R, S*** system. (Write *R* or *S* near the corresponding stereocenters)

(a)

OH
HO
NHAc
S
R
S



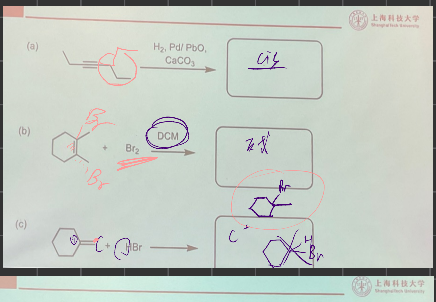
判断R，S

(b)

(c)

(d)

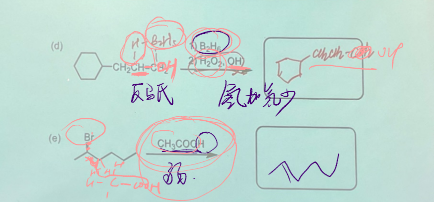
COOH
H
CH3
H
COOH
H OH
H OH
COOH
R
Cl
H
H
Br
S



(a) H_2, Pd/ PbO, $CaCO_3$ → cis

(b) + Br_2 DCM → 反式

(c) + HBr →

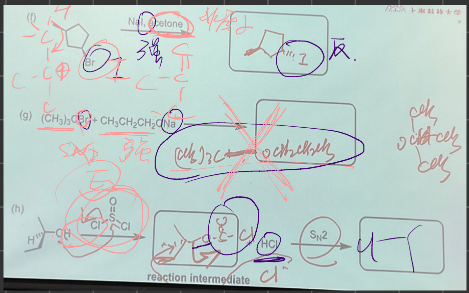


(d) $-CH_2CH=CH_2$ 1) B_2H_6 2) H_2O_2, OH^- →

反马氏 氢加氢少

(e) Br CH_3COOH →

CH2CH2CH2OH

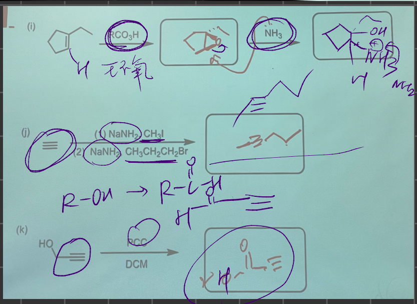


(f) NaI, acetone → 反.

(g) $(CH_3)_3CBr$ + $CH_3CH_2CH_2ONa$ →

(h) → reaction intermediate → S_N2 →

Br
I
I
SN2
(CH3)3C–OCH2CH2CH3
O
Cl S Cl
H OH
HCl
Cl⁻



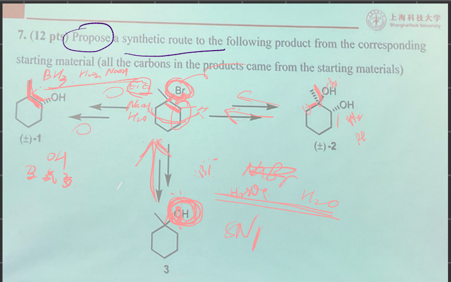
(i) RCO_3H → NH_3 →

环氧

(j) (1) $NaNH_2$, CH_3I (2) $NaNH_2$, $CH_3CH_2CH_2Br$ →

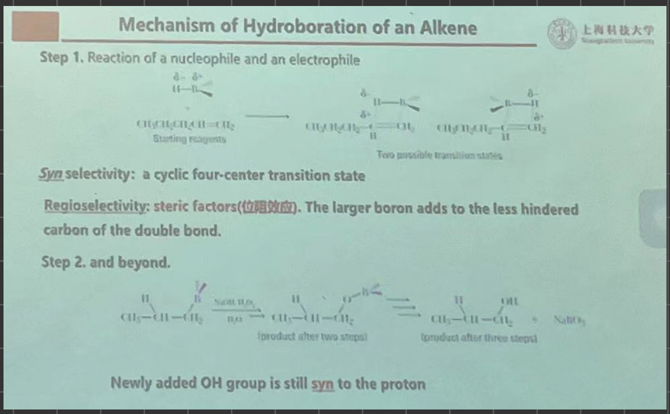
R–OH → R–C(=O)H

(k) HO PCC DCM →



7. (12 pts) Propose a synthetic route to the following product from the corresponding starting material (all the carbons in the products came from the starting materials)

BH3, H2O2, NaOH
OH
Br
NaOH H2O
OH
OH
(±)-1
(±)-2
OH
3
SN1
H2O



Mechanism of Hydroboration of an Alkene

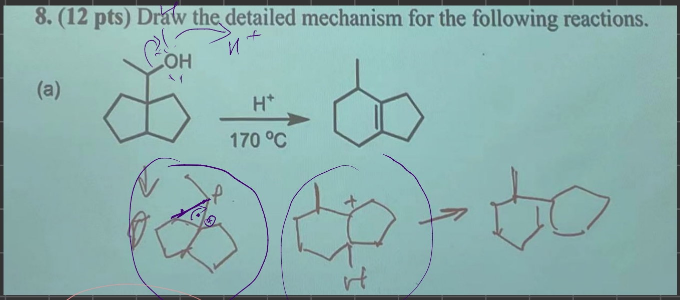
Step 1. Reaction of a nucleophile and an electrophile

Syn selectivity: a cyclic four-center transition state

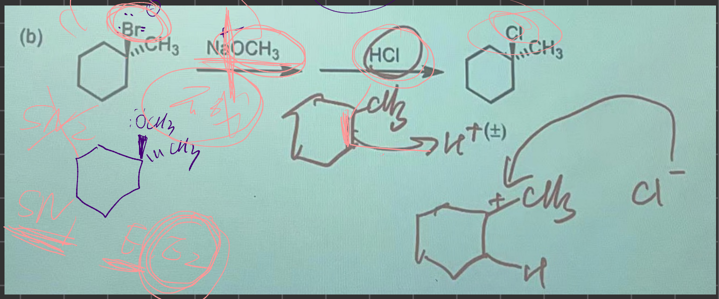
Regioselectivity: steric factors(位阻效应). The larger boron adds to the less hindered carbon of the double bond.

Step 2. and beyond.

Newly added OH group is still syn to the proton



8. (12 pts) Draw the detailed mechanism for the following reactions.



-OH 多 溶解度

1级醇 酸性 max

极性大

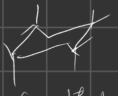
H 25

1° 命名 5~6题　英文命名 画结构
common name　溶剂、试剂
PPT 出现过的

2° 给几个化合物，比较酸/碱，排序 最酸最碱
C⁺离子重排，最稳定
醇溶解度比较

3° Newman 投影　最稳定/不稳定

4° 手性 R/S　[illegible] 内消旋　messal compound
怎么判断.



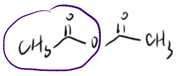
怎么画
横着更稳定.

5° 反应　有机反应在…情况下反应.　反应条件/反应产物
SN1 SN2 E1 E2　醇/环氧

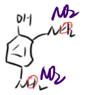
6° 反应机理　电子怎么流 ⟶ 有起始，有末态
12分一个题

1. (10 pts) Draw the structures for the following corresponding names:

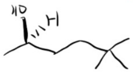
a) acetic anhydride 酸酐



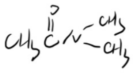
b) 2,6-Dinitrophenol



c) (*R*)-5,5-dimethyl-2-hexanol 醇 必考



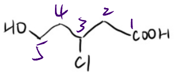
d) *N,N*-Dimethylacetamide



酚 醇 氨 羧酸

酯 醚

e) 3-chloro-5-hydroxypentanoic acid



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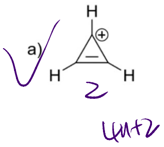
2. **(8 pts)** Please answer the following questions as shown below.
(a) Which ones of the following compounds are aromatic?

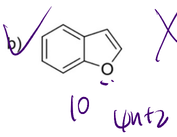
芳香性

C+ 旁边不能双键

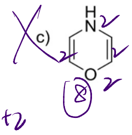
a)



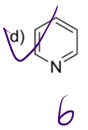
b)



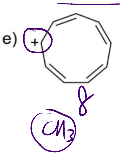
c)



d)



e)



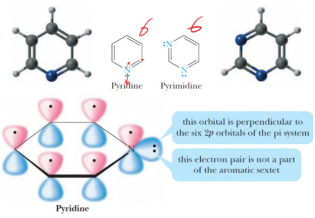
Aromaticity(芳香性)

Hückel' s rule(休克规则):

1. Have one 2*p* orbital on each of its atoms.
2. Be **planar** or nearly planar, so that there is continuous overlap all 2*p* orbitals of the ring.
3. Have 2, 6, 10, 14, 18, (**4n+2**) pi electrons in the cyclic arrangement of 2*p* orbitals.

Heterocyclic compound(杂环化合物): contains one or more atoms other than carbon in its ring.

*Both are **aromatic**!*



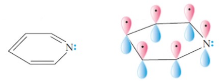
6
6
Pyridine
Pyrimidine
this orbital is perpendicular to the six 2p orbitals of the pi system
this electron pair is not a part of the aromatic sextet
Pyridine

In pyridine(吡啶), nitrogen is ***sp*² hybridized**. Only **one electron in unhybridized p** orbital

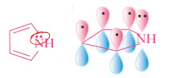
---

A Lone Pair of Electrons in an Aromatic Pi System

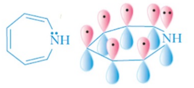
(a) Determine whether the atom containing the lone pair of electrons is part of a double bond.



(b) If it is not part of a double bond, it is **possible** for them to be part of the pi system.

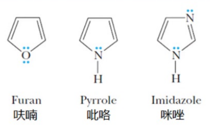


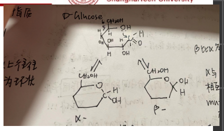
The lone pair on nitrogen gives the pi system six electrons. Therefore, the nitrogen should be *sp*² hybridized.

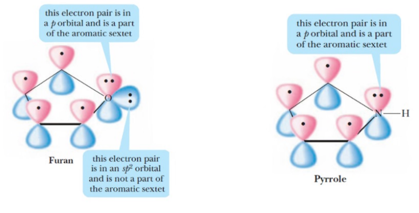


The lone pair gives the pi system eight electrons. Therefore, the nitrogen should not be *sp*² hybridized.

Aromatic !







this electron pair is in a *p* orbital and is a part of the aromatic sextet
Furan
this electron pair is in an *sp*² orbital and is not a part of the aromatic sextet
this electron pair is in a *p* orbital and is a part of the aromatic sextet
Pyrrole

Heteroatom is *sp*² hybridized, and its unhybridized 2*p* orbital contain 2 electrons

---

Leaving group

酸碱

pH

没有common name

(b) Arrange the following groups in the order of increased leaving ability.

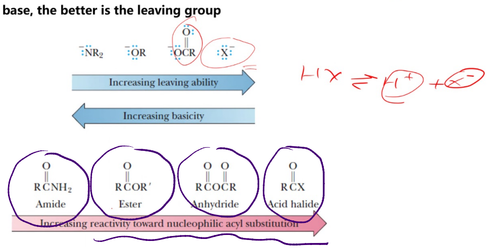
H_2O OH^- OAc^- I^-

$OH^- < OAc^- < H_2O < I^-$

考过

Nucleophilic acyl substitution

- The weaker the base, the better is the leaving group



- Acid halides and acid anhydrides are so reactive that they are not found in nature

考察化学位移 振动图.

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3. **(9 pts)** Three compounds (**A, B** and **C**) have the same molecular formula $C_4H_8O_2$. Their ^{13}C NMR of these three compounds all have obvious signal at δ 160-180. 不饱和度
Their ^1H NMR spectrum data are given below:
A, ^1H NMR 11.00 (s, 1H), 2.34 (t, 2H), 1.65 (m, 2H), 0.98 (t, 3H).
B, ^1H NMR 3.71 (s, 3H), 2.42 (q, 2H), 1.12 (t, 3H). 4重峰 $CH_3CH_2-\overset{O}{C}-OCH_3$
C, ^1H NMR 4.12 (q, 2H), 2.05 (s, 3H), 1.26 (t, 3H). $CH_3CH_2-O\overset{O}{C}CH_3$
1) Calculate the **index of hydrogen deficiency** for compound **A, B** and **C**.
2) Deduce the structure of compound **A, B** and **C**.
3) What's the most notable differences of the IR spectrum of compound **A** when compared with compound **B** and **C**?

1740 -1800 C=O双键

$\frac{2n+2-8}{2}$ =1

羧酸

红外最大的区别

H谱 C谱

→ 160-180 是什么?

考你的H谱. 红外. C谱 振动图

A $CH_3CH_2CH_2COOH$

B $CH_3-O-CH-\cancel{CH-CH_3}$ $CH_3-O\overset{O}{C}CH_2CH_3$

C $CH_3CH_2O\overset{O}{C}CH_3$

2) 3400 宽峰.

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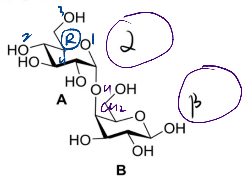
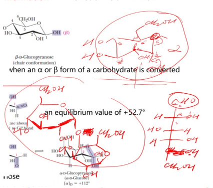
4. **(8 pts)** The structure of a disaccharide (二糖) is shown below.

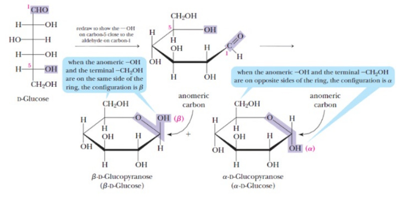




nformation in which strain is a minimum
β-D-Glucopyranose (chair conformation)
when an α or β form of a carbohydrate is converted
an equilibrium value of +52.7°
α-D-Glucopyranose
CHO

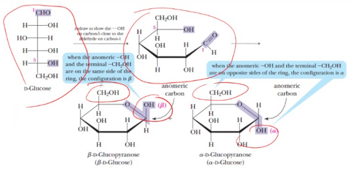
1) Indicate the configuration of the linkage between these two saccharides (α- or β-configuration?).
2) Is this disaccharide a reducing sugar?
3) Draw the **Haworth projection** of this disaccharide.
4) Draw the **Fisher project** of each monosaccharide (单糖) **A** and **B**.

Haworth projection



CHO
D-Glucose
redraw to show the —OH on carbon-5 close to the aldehyde on carbon-1
when the anomeric —OH and the terminal —CH2OH are on the same side of the ring, the configuration is β
anomeric carbon
when the anomeric —OH and the terminal —CH2OH are on opposite sides of the ring, the configuration is α
β-D-Glucopyranose (β-D-Glucose)
α-D-Glucopyranose (α-D-Glucose)

Haworth projection


D-Glucose
anomeric carbon
β-D-Glucopyranose (β-D-Glucose)
α-D-Glucopyranose (α-D-Glucose)
when the anomeric –OH and the terminal –CH₂OH are on the same side of the ring, the configuration is β
when the anomeric –OH and the terminal –CH₂OH are on opposite sides of the ring, the configuration is α

The ring is drawn flat and viewed through its edge

1) Anomeric carbon are written at the **right** and the hemiacetal oxygen at the **back right**

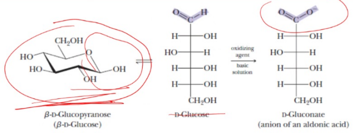
2) Groups on the **right** in the Fischer projection **point down** in the Haworth projection,

3) Groups on the **left** in the Fischer projection **point up** in the Haworth projection.

4) For a **D-monosaccharide**, the terminal -CH_2OH **points up** in the Haworth projection.

5) If the anomeric -OH group is on **the same side** as the terminal -CH_2OH, its configuration is **β**; if the anomeric -OH group is on the **opposite side**, it is **α**.

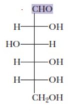
Oxidation to Aldonic Acids(醛糖酸 , Reducing Sugars)


β-D-Glucopyranose (β-D-Glucose)
D-Glucose
oxidizing agent
basic solution
D-Gluconate (anion of an aldonic acid)

Reducing sugar A carbohydrate that reacts with an oxidizing agent to form an aldonic acid.

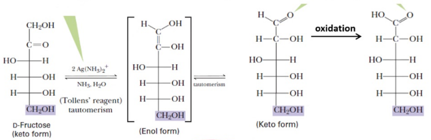
Determine If a Carbohydrate Is a Reducing Sugar

1) Any carbohydrate that is an aldehyde is a reducing sugar.

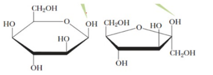

D-Glucose

Oxidation to Aldonic Acids(醛糖酸 , Reducing Sugars)

2) Any **ketose** that is in equilibrium with its aldehyde is a reducing sugar.



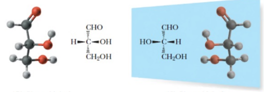
3) Any ring carbohydrate that exists as a **hemiacetal** will be in equilibrium with its acyclic form.

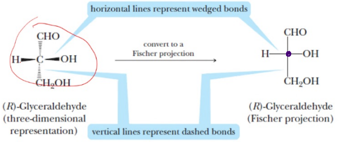


aldose or ketose

Stereoisomerism of monosaccharide

Fischer Projection Formulas (Fisher投影式)


(R)-Glyceraldehyde (S)-Glyceraldehyde


horizontal lines represent wedged bonds
convert to a Fischer projection
vertical lines represent dashed bonds
(R)-Glyceraldehyde (three-dimensional representation)

(R)-Glyceraldehyde (Fischer projection)

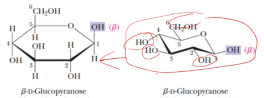
1) the most oxidized carbon toward the top

2) horizontal lines represent bonds **projecting forward** from the stereocenter, whereas **vertical lines** represent bonds **projecting to the rear**. **（横前竖后）**

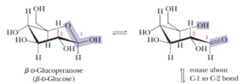
Conformation Representations

Pyranoses is more accurately represented as a **chair conformation** in which strain is a minimum

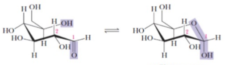
需掌握!


β-D-Glucopyranose (Haworth projection) β-D-Glucopyranose (chair conformation)

Mutarotation (变旋) The change in optical activity that occurs when an α or β form of a carbohydrate is converted to an **equilibrium mixture** of the two forms.


β-D-Glucopyranose (β-D-Glucose) $[\alpha]_D = +18.7°$

an equilibrium value of +52.7°

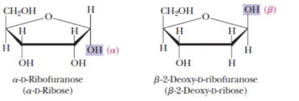


α-D-Glucopyranose (α-D-Glucose) $[\alpha]_D = +112°$

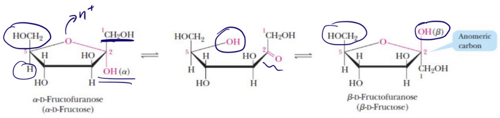
64% β-D-glucopyranose and 36% α-D-glucopyranose

β-D-glucopyranose is more stable than the α-anomer

Haworth projection


α-D-Ribofuranose (α-D-Ribose) β-2-Deoxy-D-ribofuranose (β-2-Deoxy-D-ribose)

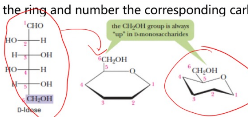
Fructose(果糖)



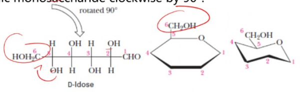
α-D-Fructofuranose (α-D-Fructose) β-D-Fructofuranose (β-D-Fructose)

Determine Stereochemistry of OH Groups in Cyclic Monosaccharides

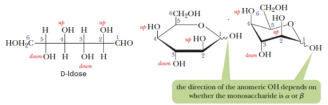
a) Draw the Haworth or chair form of the ring and number the corresponding carbon atoms



b) Rotate the drawing of the acyclic monosaccharide clockwise by 90°:



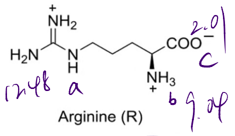
c) The direction (up or down) of the OH group on the rotated structure is the direction the OH group in on the cyclic monosaccharide:


D-Idose
the direction of the anomeric OH depends on whether the monosaccharide is α or β

5. (9 pts) Arginine has the pKa values of 2.01, 9.04, 12.48.

a b c

两性化合物



Arginine (R)

1) Draw the resonance structure of the Guanidine (胍) Group of Arginine (精氨酸), Assign the each pK_a value of guanidine group

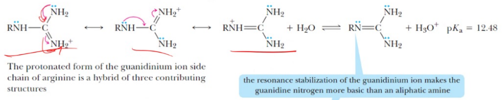
2) Calculate the PI (Isoelectric point, 等电点) of Arginine.

3) What's the net charge of Arginine at pH=8?

PI > 8 positive charge

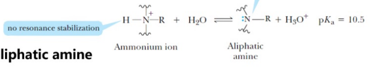
Basicity of side chains

Basicity of the Guanidine (胍) Group of Arginine (精氨酸)



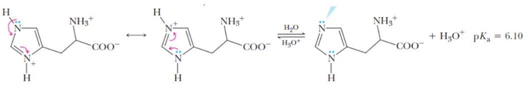
The protonated form of the guanidinium ion side chain of arginine is a hybrid of three contributing structures

the resonance stabilization of the guanidinium ion makes the guanidine nitrogen more basic than an aliphatic amine



Stronger base than an aliphatic amine

Basicity of the Imidazole (咪唑) Group of Histidine (组氨酸)



Resonance-stabilized imidazolium cation

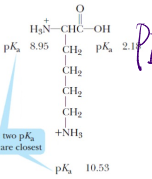
Isoelectric Point

Isoelectric point (pI, 等电点) The pH at which an amino acid, peptide, protein has no net charge.

$$\mathrm{pI} = \frac{1}{2}(\mathrm{p}K_a\ \alpha\text{-COOH} + \mathrm{p}K_a\ \alpha\text{-NH}_3^+)$$
$$= \frac{1}{2}(2.35 + 9.78) = 6.06$$

With pI for amino acids that contain either **two carboxyl or two ammonium groups**, we use **the p*K*a' s of the two groups that are closest in value**.

$$\mathrm{pI} = \tfrac{1}{2}\ (\mathrm{p}K_a\,\alpha\text{-}\overset{+}{\mathrm{N}}\mathrm{H}_3 + \mathrm{p}K_a\ \text{side chain-NH}_3^+)$$
$$= \tfrac{1}{2}\ (8.95 + 10.53) = 9.74$$



$$PI = \frac{9.04 + 12.48}{2}$$

Isoelectric Point

$$\underset{\mathrm{NH_3^+}}{\mathrm{RCHCOOH}} \underset{\mathrm{H_3O^+}}{\overset{\mathrm{OH^-}}{\rightleftharpoons}} \underset{\mathrm{NH_3^+}}{\mathrm{RCHCOO^-}} \underset{\mathrm{H_3O^+}}{\overset{\mathrm{OH^-}}{\rightleftharpoons}} \underset{\mathrm{NH_2}}{\mathrm{RCHCOO^-}}$$

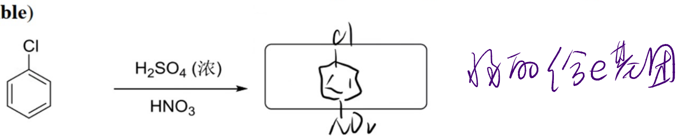
pH < pI (net charge = +) pH = pI (net charge = 0) pH > pI (net charge = −)

At pH values **greater than pI**, an increasing fraction of its molecules have a net **negative** charge.

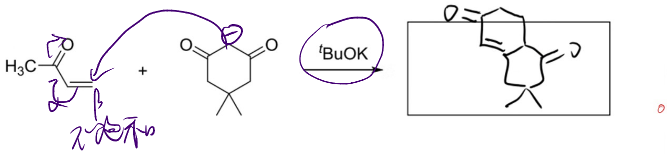
At pH values **lower than pI**, an increasing fraction of its molecules have a net **positive** charge.

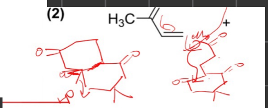
6. **(26 pts)** Fill the blanket to finish the following reactions (**more than one step is possible**)

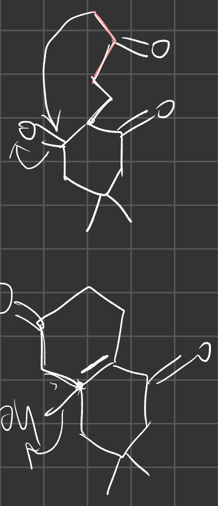
(1)



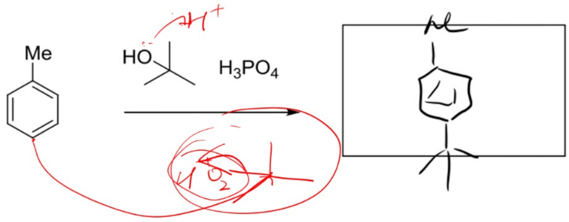
(2)



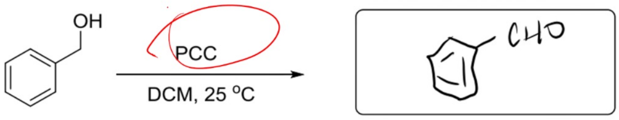




(3)

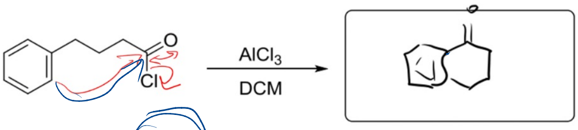


(4)



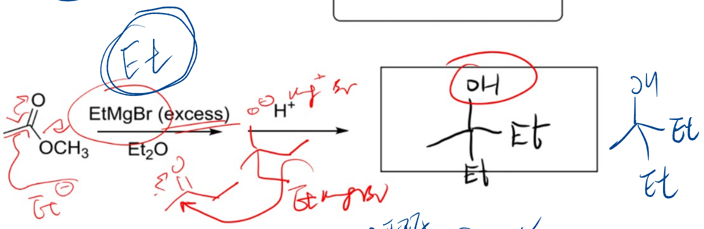
Nichoy ?

(5)

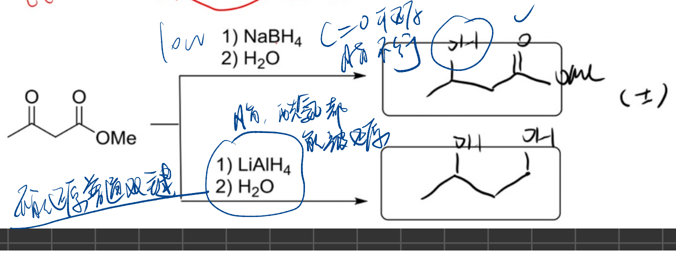


enalate

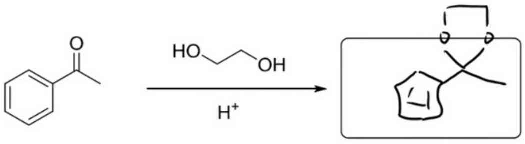
(6)



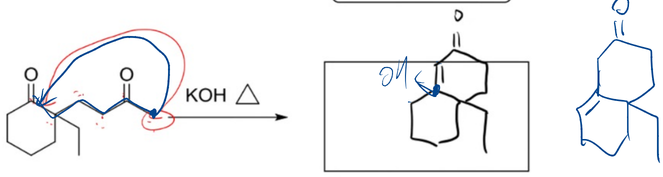
(7)



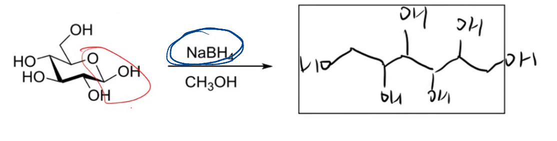
(8)



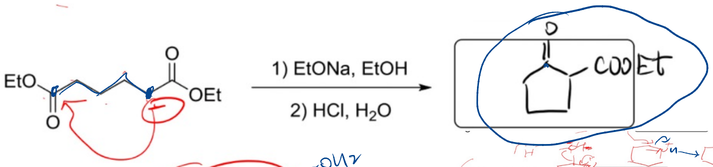
(9)

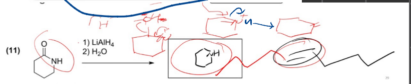


(12)



(10)





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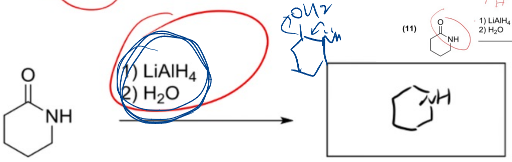
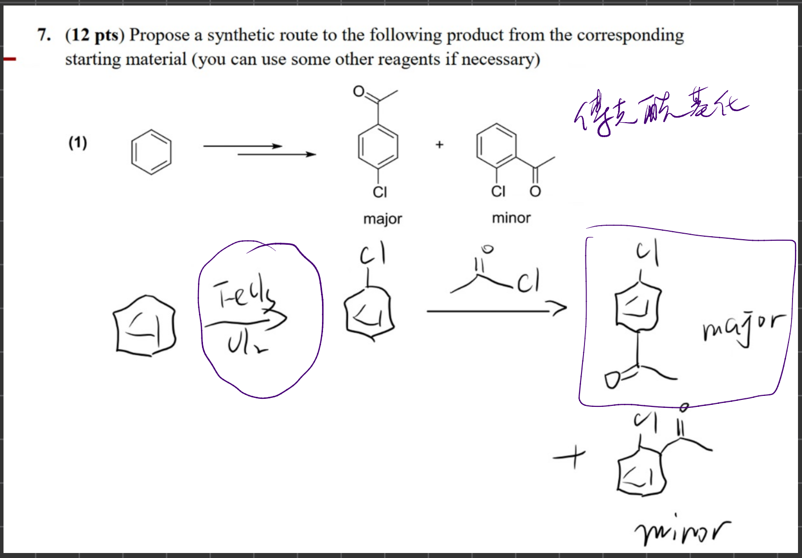




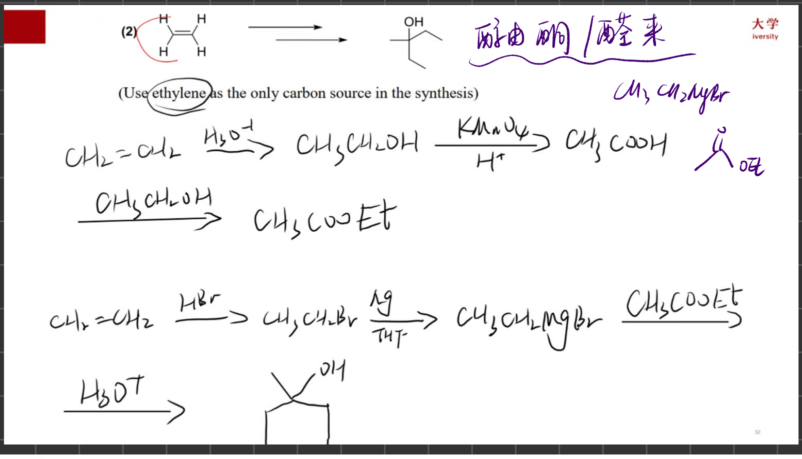
7. **(12 pts)** Propose a synthetic route to the following product from the corresponding starting material (you can use some other reagents if necessary)

(1)

major

minor

傅克酰基化



(2)

(Use ethylene as the only carbon source in the synthesis)

醇由酮/醛来

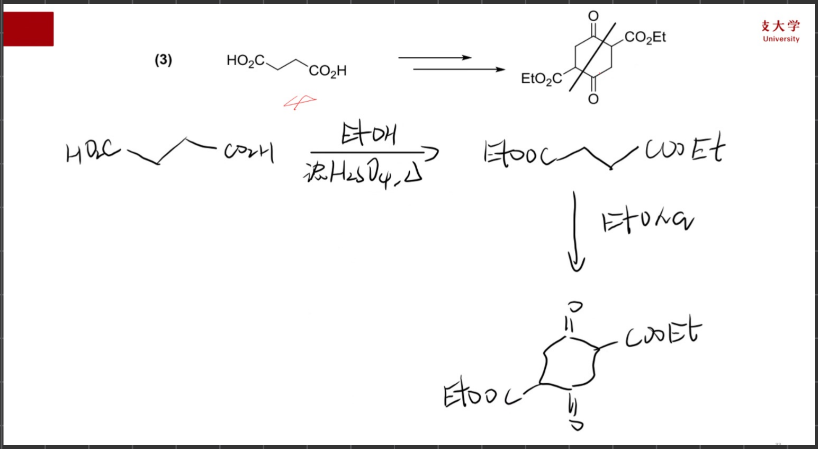
CH_3CH_2MgBr

$CH_2=CH_2 \xrightarrow{H_3O^+} CH_3CH_2OH \xrightarrow[H^+]{KMnO_4} CH_3COOH$

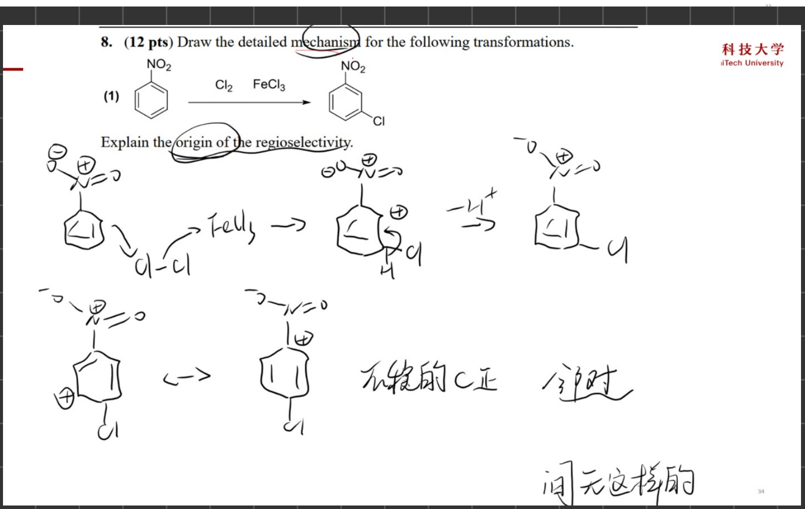
$\xrightarrow{CH_3CH_2OH} CH_3COOEt$

$CH_2=CH_2 \xrightarrow{HBr} CH_3CH_2Br \xrightarrow[THF]{Mg} CH_3CH_2MgBr \xrightarrow{CH_3COOEt}$

$\xrightarrow{H_3O^+}$



(3)



8. **(12 pts)** Draw the detailed mechanism for the following transformations.

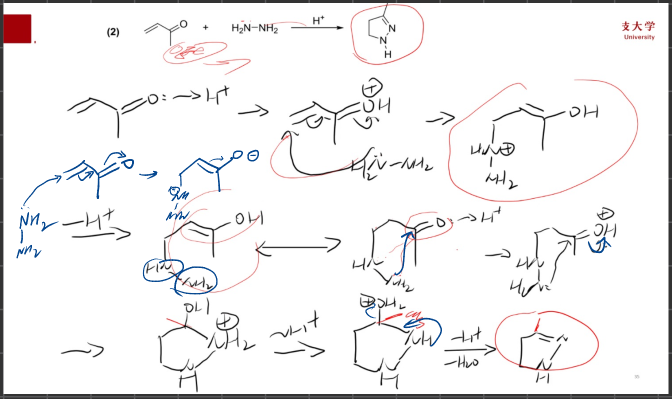
(1)

Explain the origin of the regioselectivity.

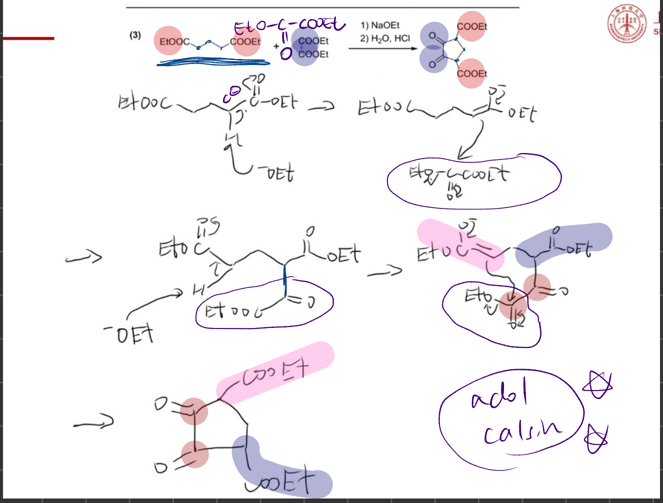
不稳的C正

邻对

间无这样的



(2)



(3)